

# NGAME Churn Persistence Index (CPI) and Dollar-Weighted CPI (wCPI): Full Derivation

Per-transaction daily measures rolled up into composite CPI / wCPI and scored against a trained baseline

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## NGAME CPI / wCPI — Full Derivation

This document gives the end-to-end computation NGAME uses for each transaction type’s daily measures — the Churn Persistence Index (CPI), its dollar-weighted variant (wCPI), the trained baseline  $(\mu, \sigma)$ , and the  $z$ -scores and sequence roll-ups that drive dashboard warnings. NGAME tracks **18 transaction types** (Customers through Vendors); pipeline output labels them  $\varphi_1 \dots \varphi_{18}$  for indexing only.

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### 1. Daily set-up — the per-type contingency counts

For each transaction type  $k$ , `NGameCpiAnalysisAgent` compares yesterday’s URI set  $Y_k$  (from `Yesterday.ttl`) against today’s URI set  $T_k$  (from `Today.ttl`).

**URI (Uniform Resource Identifier).** NGAME operates on RDF/OWL ontology snapshots in which **every individual transaction is named by a unique URI** — e.g. `qb:Invoice_4821`. The URI is the stable identity used to decide whether a transaction observed today is the *same* transaction seen yesterday. The sets  $Y_k$  and  $T_k$  are sets of transaction URIs of type  $k$ ; every count below is a count of URIs.

**QuickBooks-specific caveat.** The metric is only as reliable as QuickBooks’ URI assignment. QuickBooks mints a new identifier when a transaction is *deleted and recreated* (or voided and re-entered) rather than edited in place. The original URI is counted in  $b_k$  (removed) and the replacement in  $c_k$  (added) — registering as churn even when the transaction is economically unchanged. An *in-place* edit (amount or memo changed without a new URI) leaves set membership intact, so count-based  $\text{CPI}_k$  may not move; such edits can surface through dollar-weighted  $\text{wCPI}_k$  (§3). The AdditionRate / DeletionRate split (§2) distinguishes directional churn (additions vs. removals).

| Symbol | Definition            | Meaning                                     |
|--------|-----------------------|---------------------------------------------|
| $a_k$  | $ Y_k \cap T_k $      | Present in <b>both</b> snapshots (retained) |
| $b_k$  | $ Y_k \setminus T_k $ | Yesterday <b>only</b> (removed)             |
| $c_k$  | $ T_k \setminus Y_k $ | Today <b>only</b> (new additions)           |
| $n_k$  | $a_k + b_k + c_k$     | Total unique transactions across both days  |

Cell  $d$  (transactions absent from both days) is **not used**: the universe of non-occurring transactions is undefined for financial ledger data.

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## 2. Daily per-type measures — CPI and directional rates

The **Churn Persistence Index** is the **Jaccard similarity** of the two URI populations, with two directional companions that sum to 1:

$$\boxed{\text{CPI}_k = \frac{a_k}{a_k + b_k + c_k} = \frac{a_k}{n_k}}$$

$$\text{AdditionRate}_k = \frac{c_k}{n_k}, \quad \text{DeletionRate}_k = \frac{b_k}{n_k},$$

$$\text{CPI}_k + \text{AdditionRate}_k + \text{DeletionRate}_k = 1.0.$$

If  $n_k = 0$  (no transactions of type  $k$  in either snapshot), then  $\text{CPI}_k = 1.0$  by convention (“absence is stable”).

The triplet  $(\text{CPI}_k, \text{AdditionRate}_k, \text{DeletionRate}_k)$  characterises the day’s dynamics for type  $k$ . A **low** CPI (high churn) is the primary fraud concern; a **high** AdditionRate flags bulk insertion; a **high** DeletionRate flags retroactive deletion.

### Worked example — single-day CPI

**Customers: one record added.**

$$\text{Before: } a = 50, b = 0, c = 0 \Rightarrow \text{CPI} = \frac{50}{50} = 1.000$$

$$\text{After: } a = 50, b = 0, c = 1 \Rightarrow \text{CPI} = \frac{50}{51} = 0.980$$

**Partial churn (illustrative),  $a = 2, b = 2, c = 1, n = 5$ :**

$$\text{CPI} = \frac{2}{5} = 0.400, \quad \text{DeletionRate} = \frac{2}{5} = 0.400, \quad \text{AdditionRate} = \frac{1}{5} = 0.200,$$

with  $\text{CPI} + \text{AdditionRate} + \text{DeletionRate} = 1.000$ .

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## 3. Dollar-weighted CPI (wCPI)

Unweighted CPI treats a \$50 expense identically to a \$500{,}000 wire. The weighted form keeps the Jaccard structure but replaces set cardinality with the sum of transaction amounts  $w_i$  (from `qb:hasTotalAmount`; entity-type records with no dollar amount default to  $w_i = \$1$ ):

$$\text{wCPI}_k = \frac{\sum_{i \in Y_k \cap T_k} w_i}{\sum_{i \in Y_k \cup T_k} w_i} = \frac{\text{retained dollars}}{\text{total dollars in play}}.$$

Retained URIs use **today’s** amount; removed URIs use **yesterday’s** amount.

### Worked example — a \$50,000 expense hidden in a normal day

Twenty transactions: 19 of \$50 retained, plus one new \$50{,}000 expense ( $a = 19, b = 0, c = 1$ ).

$$\text{Unweighted: } \text{CPI} = \frac{19}{20} = 0.9500 \quad (\text{looks normal})$$

$$\text{Weighted: } \text{retained } \$ = 19 \times \$50 = \$950, \quad \text{union } \$ = \$950 + \$50,000 = \$50,950,$$

$$\text{wCPI} = \frac{950}{50,950} = 0.0186 \quad (\text{alarm fires}),$$

$$\text{Divergence} = |\text{CPI} - \text{wCPI}| = |0.9500 - 0.0186| = 0.931 \rightarrow \mathbf{HIGH}.$$

| Metric               | Value  | Reading                                |
|----------------------|--------|----------------------------------------|
| Unweighted CPI       | 0.9500 | Looks normal (19 of 20 URIs retained)  |
| Dollar-weighted wCPI | 0.0186 | Exposure is overwhelmingly new dollars |
| Divergence           | 0.931  | <b>HIGH</b>                            |

Dollar-alarm thresholds on divergence:  $> 0.30 = \text{HIGH}$ ,  $> 0.15 = \text{MEDIUM}$ . A parallel **dollar  $z$ -score** (today's wCPI vs. trained  $\mu_w, \sigma_w$ ) is also computed when wCPI baselines exist in the matrix.

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## 4. The trained baseline — daily $\mu$ and $\sigma$ per type

Over an  $N$ -day training window, each type accumulates a distribution of daily CPI values. The Training Matrix stores **Mean CPI** ( $\mu$ ) and **Std Dev CPI** ( $\sigma$ ) per row (columns B and C), plus daily CPI values in columns Day 1 ... Day  $N$ .

$$\mu_k = \frac{1}{N} \sum_{t=1}^N \text{CPI}_k^{(t)}, \quad \sigma_k = \sqrt{\frac{1}{N-1} \sum_{t=1}^N \left( \text{CPI}_k^{(t)} - \mu_k \right)^2}.$$

A separate **wCPI\_Baseline** sheet stores  $(\mu_w, \sigma_w)$  for dollar-weighted CPI. Optional day-of-week stratification computes separate  $(\mu_k, \sigma_k)$  per weekday (requires roughly 60–90 training days for a stable  $\sigma$ ).

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## 5. Daily anomaly score — $z$ against the trained distribution

NGameChurnComparisonAgent compares today's CPI array to the matrix baseline:

$$z_k = \frac{\text{CPI}_k^{\text{today}} - \mu_k}{\sigma_k}.$$

| Condition              | Level  | Statistical meaning                             |
|------------------------|--------|-------------------------------------------------|
| $ z_k  \leq 2.0$       | LOW    | Within roughly 95% of training distribution     |
| $2.0 <  z_k  \leq 3.0$ | MEDIUM | Outside 95%, within 99.7%                       |
| $ z_k  > 3.0$          | HIGH   | Beyond $3\sigma$ — rare under normal operations |

**Percent deviation** (secondary dashboard readout):

$$\text{percent\_deviation}_k = \frac{|\text{CPI}_k^{\text{today}} - \mu_k|}{\mu_k} \times 100.$$

**Composite alarm** fires when deviation level is HIGH or MEDIUM, or when the dollar signal (divergence or dollar  $z$ -score) is elevated.

#### Worked example — ChartOfAccounts

With today's CPI = 0.065 against trained  $\mu = 0.049$ ,  $\sigma = 0.004$ :

$$z = \frac{0.065 - 0.049}{0.004} = 4.0 \rightarrow \mathbf{HIGH}.$$

## 6. Composite roll-up — multi-day sequence (SCPI / weighted SCPI)

A slow-burn pattern spreads small daily moves that never trip a single-day alarm. `NGameSequenceCpiAgent` computes the **geometric mean** of the last  $L$  daily CPIs (default  $L = 5$ ), scored with a  $\sqrt{L}$ -tightened  $z$ :

$$\text{SCPI}_k(t, L) = \left( \prod_{i=1}^L \text{CPI}_k(t - i + 1) \right)^{1/L},$$

$$z_k^{\text{seq}} = \frac{\text{SCPI}_k - \mu_k}{\sigma_k / \sqrt{L}}.$$

The same  $\mu_k, \sigma_k$  from the Training Matrix are used — no schema change. Weighted SCPI applies the same construction over the daily wCPI buffer.

A type drifting at only  $-2\sigma/\text{day}$  (MEDIUM at most on any single day) can score  $z^{\text{seq}} \approx -4.47$  on day 5 — a HIGH alert the per-day detector would not raise alone.

The rolling buffer (`NGAME_CPI_Rolling_Buffer.json`) stores up to 30 days of 18-element CPI (and wCPI) arrays; the sequence path returns `ready=False` until  $L$  days exist.

## 7. Core implementation (`ngame_cpi_analysis_agent.py`)

Per-type CPI from `_calculate_cpi_for_transaction_type`:

```
a = len(yesterday_uris & today_uris)    # retained
b = len(yesterday_uris - today_uris)    # removed
c = len(today_uris - yesterday_uris)    # added
n = a + b + c                          # total unique
```

```

if n == 0:
    # stable by convention
    return {'cpi': 1.0, 'addition_rate': 0.0, 'deletion_rate': 0.0,
            'retained': 0, 'added': 0, 'removed': 0}

return {
    'cpi': a / n,
    'addition_rate': c / n,
    'deletion_rate': b / n,
    'retained': a, 'added': c, 'removed': b,
}

```

Dollar-weighted CPI: `_calculate_weighted_cpi_for_transaction_type`. Comparison and alarms: `ngame_churn_comparison_agent.py`. Sequence path: `ngame_sequence_cpi_agent.py`.

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*Verified against NGAME POC codebase and NGAME\_Training\_Matrix.xlsx (Training Matrix + wCPI\_Baseline sheets). The \$50{,}000 wCPI example is from agent test W2.*